

When two mature individuals (of opposite sex) participate in the reproductive process involving fusion of male and female gametes (**the products of meiosis** usually), it is called **sexual reproduction**. These haploid gametes fuse to form the diploid zygote, which develops into the new organism. It is an elaborate, complex and slow process compared to asexual reproduction. Because of the fusion of male and female gametes, sexual reproduction results in offspring that are not identical to the parents or amongst themselves.

5.4.1 Sexual reproduction in Protozoans

It takes place by the fusion of 'pronuclei' (haploid gametic nuclei) with or without the formation of gametes. Genetic recombination occurs in sexual reproduction during the formation of pronuclei or gametic nuclei (due to **meiotic crossovers**) and also due to the fusion of gametes from two different individuals. Sexual reproduction in protozoans occurs by syngamy. Conjugation, 'a process of nuclear reorganisation that occurs in some ciliates' also brings about a kind of sexual multiplication of the organism.

5.4.2 Syngamy

It is the fusion of two gametes.

- i) The fusion of similar gametes is called **isogamy** e.g. *Monocystis*.
- ii) The fusion of dissimilar gametes is called '**anisogamy**' e.g. *Plasmodium*.
- iii) **Hologamy** is the fusion of two mature organisms which do not form gametes but behave as gametes. e.g. *Trichonympha*.

* Union of pronuclei of the gametes is called '**amphimixis**' and the resultant nucleus is called **synkaryon**.

5.4.3 Conjugation

It is observed in ciliate protozoans such as *Paramecium* and *Vorticella*. Wichterman defined conjugation as a temporary union between **two senile** ciliates that belong to two different 'mating types' for the exchange of nuclear material and its reorganization. It restores the **vigour** and **vitality** which are lost due to chromosomal imbalance in their macronuclei caused by repeated **amitotic** divisions. Unfavourable conditions also induce conjugation.

GLOSSARY

Phototropism: It is the tendency of an organism to move towards or to move away from the source of light stimulus.

Organelle: A living component of the cell with a specific structure and function. e.g., Mitochondrion, Lysosome, Ribosome, Golgicomplex, etc.

Heliopodia: These are sunray-like pseudopodia present all over the surface of the body, as in 'sun animal cules' (*Actinophrys*, *Actionsphaerium*).

Tubulin: A protein with which the microtubules of cytoskeleton are formed.

Kinetosome: It is a modified centriole from which the cilium or flagellum is formed.

Suctoria: Advanced ciliate protozoans, which possess suctorial tentacles in the adult condition and cilia in the juvenile stages. e.g. *Acineta*.

Senile: Becoming old.

Nexin links: Proteins that connect the 'A' tubule of one doublet with the 'B' tubule of the next doublet in a clockwise manner in an axoneme are called nexin links. They maintain the integrity of the axoneme by holding doublets together.

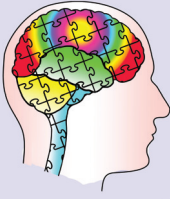
VERY SHORT ANSWER TYPE QUESTIONS

1. Draw a labelled diagram of T.S of flagellum.
2. List any two differences between a flagellum and a cilium.
3. What are dynein arms? What is their significance?
4. What is a kinety?
5. Distinguish between synchronous and metachronous movements.

6. Why do we refer to the offspring, formed by asexual method of reproduction, a clone?
7. Distinguish between proter and opisthe.
8. How is sexual reproduction advantageous in evolution?
9. Distinguish between lobopodium and filopodium. Give an example to each of them.
10. Define conjugation with reference to ciliates. Give two examples.

SHORT ANSWER TYPE QUESTIONS

1. Name the system that controls the fastest swimming movement of protozoans and write its components.
2. Write the mechanism of bending of flagellum and explain effective and recovery strokes.
3. What are lateral appendages? Based on their presence and absence, write the various types of flagella giving at least one example for each type.
4. Describe the process of transverse binary fission in *Paramecium*.
5. Describe the process of longitudinal binary fission in *Euglena*.
6. Write a short note on multiple fission.
7. Give an account of pseudopodia.
8. Give an account of the ultra structure of an axoneme.
9. Draw a neat labelled diagram of ***Euglena***.
10. Draw a neat diagram of ***Paramecium*** and label its important structures/ components.



For ignited minds

- NOT FOR EVALUATION

1. If, a row of lateral appendages is added to the flagellum of *Chilomonas*, what type of flagellum does it transform into, hypothetically speaking?
2. What are the structures associated with locomotion that occur in higher animals such as man?
3. What is the primary source of energy for any type of movement, be it in lower organism or higher organism?
4. Why should conjugation occur in ciliates only, why not in other protozoans. **(CLUE)**: In the heteronuclear organisms, the vegetative nucleus divides **amitotically**.
5. If the dynein arms are immobilized, do flagella or cilia or both stop functioning?
6. What are the two structures which fix the doublets of a cilium/flagellum and keep them in position or in other words maintain the integrity of microtubules?
7. There is some indication concerning certain gametes of higher organisms, that at least indicate the origin of higher animals from some simple protists. **Can you explain this concept?**
8. *Euglena* has chromatophores. Yet zoologists consider it an animal. From the few things you have learnt in this lesson about *Euglena*, what features speak in favour of the view of zoologists?
9. A *Paramecium* can produce more daughter paramecia at the end of conjugation. The same *Paramecium* can produce a clone of daughter paramecia through multiplication by binary fission. Is there any difference between the two types of daughter paramecia? Discuss with your teacher and find whether your answer is correct or not?
[CLUE- Production of a clone does not involve pronuclei].
10. All undulipodia (**Hyman's terminology**) do not show undular movement. Do you agree with the statement. If so, how?
11. The total number of microtubules present above the level of pellicle of a flagellum is 20.
12. The total number of microtubules present in the kinetosome is 27 (3×9)
13. The 9 **radial spokes** in an axoneme prevent undue sliding of peripheral doublets.
14. The tubules that connect kinetosome to the plasma membrane and nucleus are called **rootlets**.
15. The flagellate with highest number of flagella is *Trichonympha*.
16. Cilia and flagella arise from kinetosomes, their **progenitors**.
17. Flagella and cilia are considered **homologous** and **analogous organelles**.
18. The dynein arms are the sites of activity of **ATPase**, which releases energy from ATP.
19. **Uniparental** inheritance is involved in asexual reproduction, whereas **biparental** inheritance is involved in sexual reproduction.
20. The first stage in the life of a sexually reproducing organism is a **totipotent** diploid cell, the **zygote**.